# EXACT ERROR COVARIANCES FOR BLOCK FLOATING POINT REDUCTIONS

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**Abstract.** Selecting a shared scale from a block maximum couples the quantization errors. We derive exact finite-block error moments from one-dimensional truncated distributions, including continuous laws and atoms. For identically distributed independent inputs, the conditional pair covariance is nonpositive; variation of the conditional mean supplies a competing positive contribution. The resulting covariance gives the variance and mean-square error of weighted reductions. Dropping the conditional pair correction while retaining the conditional moments overestimates sum-variance inflation by at most one. The formulas extend to independent heterogeneous inputs and, by conditioning, to correlated inputs with latent factors. An exponential-input theorem shows that variance inflation can diverge along a dyadic subsequence despite vanishing pair correlation and finite moments of every order. Rational enumeration, high-precision integration, and Monte Carlo validate the moment calculations. Two-dimensional elliptic experiments connect the moments to solution energy error, spatial block layout, and precision allocation at a fixed storage budget. At seven bits per source value, spatial precision placement reduces measured RMS error by 22.4% relative to uniform placement; a uniform-noise model selects a codec with 4.1% higher error than the best tested candidate. Correlated-source experiments retain cross-block dependence through the latent-factor mixture. Bit-packing audits and conjugate-gradient solves verify the format and predicted input-conversion errors. An independent train/validation experiment tests allocation after estimating the source law from data.

**Key words.** block floating point, quantization, rounding error, covariance, weighted reduction, stochastic rounding

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## 1. Introduction.

Block floating point represents a vector by integer coefficients and a common power-of-two scale. The scale is usually selected from the block maximum, so it depends on the same data that are subsequently rounded. This choice couples the rounding errors. In a sum or a matrix–vector product, even a small covariance between two coordinates can matter because it contributes once for every coordinate pair. Predicting the variance of an individual error is therefore insufficient for predicting the error of a reduction.

Classical statistical quantization models approximate residuals by centered uniform noise [17, 18]. Block floating point roundoff has a long history [10], and statistical noise models continue to inform accelerator design [11, 16]. Microscaling formats add a contemporary motivation [12]. The shared-scale dependence is also recognized: Rohani and Fakhraie [13] discuss correlations in dynamically rescaled structures, and Soloveychik et al. [15] use maximum order statistics to derive inner-product error bounds and study block-size selection for Gaussian data. The latter model approximates nonmaximum residuals by uniform noise. Fox and Lindstrom [6] analyze systematic bias in the ZFP compression pipeline. These studies leave a finite-block question: how can one compute the error covariance of a maximum-selected quantizer directly from the input law?

We derive such a calculation from one-dimensional truncated moments. For independent, identically distributed inputs, conditioning on a binade of the maximum excludes the event that every coordinate lies below its lower endpoint. This exclusion gives an explicit nonpositive conditional covariance. Variation of the conditional

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error mean across binades supplies a competing nonnegative contribution. Their sum determines whether a reduction is inflated or attenuated. The calculation also extends to independent inputs with different distributions, for which the conditional pair correction can have either sign. Conditioning on latent factors then provides moments for correlated inputs and covariances between blocks.

The covariance matrix gives exact variance and mean-square error formulas for deterministic weighted reductions. It also provides a useful approximation guarantee: retaining the conditional moments while dropping the negative pair correction overestimates the sum's variance inflation factor by at most one. Thus the approximation becomes relatively accurate when inflation is large, while the full formula resolves small inflation and deflation. The complete covariance matrix supplies the moments needed for propagation through a linear operator.

The large-block limit reveals a separate obstruction. Bounded support with a nondegenerate limiting error gives unit inflation. With unbounded support, the pair correlation tends to zero, yet the number of pairs can amplify it. An exponential-input counterexample proves that inflation diverges along a dyadic subsequence even though every input moment is finite. A decreasing pair correlation therefore cannot justify an independent-error model for large reductions.

Our numerical study checks the formulas by exact rational enumeration, high-precision integration, and independently sampled blocks. Weighted reductions expose the role of cancellation and bias; elliptic solves connect the covariance to solution accuracy, block layout, and a fixed storage budget. A train/validation experiment tests the same allocation after the source law is estimated rather than supplied. Symmetric inputs and independently randomized stochastic rounding provide controls through their zero pair covariance [2, 3]. These results complement probabilistic analyses of arithmetic rounding [7–9], mixed-precision kernels [5], fixed-quantizer inner-product distortion [1], and state-conditioned reduction trees [14]. The contribution here is the dependence created when the input quantizer selects its own common scale.

**2. Exact finite-block analysis.** Let $X_1, \dots, X_n$ be independent copies of a real random variable $X$, where $n \ge 2$, $\mathbb{E}[X^2] < \infty$, and $L$ is a fixed nonnegative integer. Set $M_n = \max_i |X_i|$. When $M_n > 0$, define

(2.1)
$$K_n = \lfloor \log_2 M_n \rfloor, \qquad s_k = 2^{k-L}, \qquad h_k(x) = s_k \,\mathrm{RN}(x/s_k) - x, \qquad e_i = h_{K_n}(X_i),$$

with round-to-nearest, ties-to-even on an unsaturated uniform integer grid. The all-zero block has state $*$ and zero errors. The analysis treats input conversion; weights and linear operators are fixed. Independent heterogeneous laws and conditionally independent inputs are introduced in Section 2.3. For a selected nonzero block, $|X_i/s_{K_n}| < 2^{L+1}$, so the rounded integer coefficient has magnitude at most $2^{L+1}$. A signed integer of width $L + 3$ therefore suffices, including rounding to the positive endpoint. The experimental codecs use an 8-bit signed block exponent and count all coefficient and padding bits.

The error vector is exchangeable because its construction commutes with permutations of the inputs. Write

(2.2)
$$\mu = \mathbb{E}[e_1], \quad v = \mathrm{Var}(e_1) > 0, \quad \kappa = \mathrm{Cov}(e_1, e_2), \quad \mathcal{I}_n = \frac{\mathrm{Var}(\sum_i e_i)}{nv} = 1 + (n-1)\frac{\kappa}{v}.$$

The positivity of $v$ is needed only for ratios; the moment identities below also hold when $v = 0$.

**2.1. A closed conditional covariance formula.** Use the strict distribution function $G(t) = \mathbb{P}(|X| < t)$ for $t > 0$. For an exponent $k$, put

$$a_k = 2^k, \quad b_k = 2^{k+1}, \quad p_k = G(a_k), \quad q_k = G(b_k), \\ P_k = q_k^n - p_k^n, \qquad T_{\ell,k}(t) = \mathbb{E}[h_k(X)^\ell \mathbf{1}_{\{|X|<t\}}]. \tag{2.3}$$

The use of strict inequalities assigns atoms at $a_k$ and $b_k$ to the correct binades. The exponent states have total mass $1 - \mathbb{P}(X = 0)^n$; the remaining mass $P_* = \mathbb{P}(X = 0)^n$ belongs to $*$.

THEOREM 2.1 (Exact conditional moments and negative conditional covariance). *For every $k$ with $P_k > 0$, define $m_k = \mathbb{E}[e_1 \mid K_n = k]$, $t_k = \mathbb{E}[e_1^2 \mid K_n = k]$, and $c_k = \mathrm{Cov}(e_1, e_2 \mid K_n = k)$. Then*

$$m_k = \frac{q_k^{n-1} T_{1,k}(b_k) - p_k^{n-1} T_{1,k}(a_k)}{P_k}, \tag{2.4}$$

$$t_k = \frac{q_k^{n-1} T_{2,k}(b_k) - p_k^{n-1} T_{2,k}(a_k)}{P_k}, \tag{2.5}$$

$$\mathbb{E}[e_1 e_2 \mid K_n = k] = \frac{q_k^{n-2} T_{1,k}(b_k)^2 - p_k^{n-2} T_{1,k}(a_k)^2}{P_k}. \tag{2.6}$$

*If $p_k > 0$, put $u_k = T_{1,k}(a_k)/p_k$, $z_k = T_{1,k}(b_k)/q_k$, and $r_k = (p_k/q_k)^n$. Then*

$$m_k = \frac{z_k - r_k u_k}{1 - r_k}, \qquad c_k = -\frac{r_k}{(1 - r_k)^2}(z_k - u_k)^2 \le 0. \tag{2.7}$$

*Products associated with a zero-probability event in* (2.6) *are understood as zero, including when $n = 2$. If $p_k = 0$, the formulas reduce to $m_k = z_k$ and $c_k = 0$. The same identities hold for any measurable deterministic scalar response $h_k$ applied identically to each coordinate, with the required moments.*

*Proof.* The event $\{K_n = k\}$ is the difference between the nested events $\{M_n < b_k\}$ and $\{M_n < a_k\}$. Independence gives

$$\mathbb{E}[h_k(X_1)\mathbf{1}_{\{M_n<t\}}] = G(t)^{n-1} T_{1,k}(t), \\ \mathbb{E}[h_k(X_1)h_k(X_2)\mathbf{1}_{\{M_n<t\}}] = G(t)^{n-2} T_{1,k}(t)^2.$$

Subtract the expressions at $t = a_k$ from those at $t = b_k$ and divide by $P_k$. The second-moment identity follows in the same way. Substitution of $T_{1,k}(a_k) = p_k u_k$ and $T_{1,k}(b_k) = q_k z_k$ yields $\mathbb{E}[e_1 e_2 \mid K_n = k] = (z_k^2 - r_k u_k^2)/(1 - r_k)$. Subtracting $m_k^2$ gives (2.7). If $p_k = 0$, conditioning only truncates each input at $b_k$, preserving independence. ∎

The sign in (2.7) has a precise origin. Conditioning on the exponent first restricts every input to $|X_i| < b_k$, then excludes the event that all inputs satisfy $|X_i| < a_k$. The resulting selection induces a nonpositive covariance between two copies of the same scalar response. Consequently, conditional uncorrelatedness is an additional restriction, not a consequence of sharing the exponent: when $p_k > 0$, it holds exactly if $u_k = z_k$.

For the state $*$, set $m_* = t_* = c_* = 0$. Including this state in every sum, the

complete population quantities are

$$\mu = \sum_k P_k m_k, \qquad v = \sum_k P_k t_k - \mu^2, \tag{2.8}$$

$$B = \mathrm{Var}(m_{K_n}) = \sum_k P_k (m_k - \mu)^2, \qquad C = \sum_k P_k c_k \le 0, \qquad \kappa = B + C. \tag{2.9}$$

The term $B$ describes variation of the conditional mean between exponent states; $C$ is the negative correction within a state. Their sum can have either sign.

COROLLARY 2.2 (A certified bound for omitting the conditional covariance). *Let $A = \mathbb{E}[\mathrm{Var}(e_1 \mid K_n)]$, so $v = A + B$. Then*

$$\frac{nB}{v} \le \mathcal{I}_n \le 1 + (n-1)\frac{B}{v} \le n, \qquad 0 \le \left(1 + (n-1)\frac{B}{v}\right) - \mathcal{I}_n \le \frac{A}{v} \le 1. \tag{2.10}$$

*Proof.* The law of total variance gives $\mathrm{Var}(\sum_i e_i) = \mathbb{E}[\mathrm{Var}(\sum_i e_i \mid K_n)] + n^2 B \ge n^2 B$. The upper bound follows from $C \le 0$ in $\mathcal{I}_n = 1 + (n-1)(B+C)/v$; subtracting the lower bound proves the absolute-error estimate. Finally $B \le v$. ∎

Omitting $C$ therefore overestimates $\mathcal{I}_n$ by at most one. When $\mathcal{I}_n > 0$, its relative error is at most $1/\mathcal{I}_n$; the absolute bound remains valid at the positive-semidefinite boundary $\mathcal{I}_n = 0$.

**2.2. Weighted reductions and rounding controls.** The covariance matrix has the form

$$\Sigma = (v - \kappa) I + \kappa \mathbf{1}\mathbf{1}^T. \tag{2.11}$$

Its eigenvalues are $v - \kappa$ in the $n-1$ directions orthogonal to $\mathbf{1}$, and $v + (n-1)\kappa = v\mathcal{I}_n$ in the common direction. Both are nonnegative, giving the usual bounds $0 \le \mathcal{I}_n \le n$. For deterministic weights $w \in \mathbb{R}^n$, the error in an exact reduction of the quantized inputs is $E_w = \sum_i w_i e_i$. In particular,

$$\mathbb{E}[E_w] = \mu \mathbf{1}^T w, \qquad \mathbb{E}[E_w^2] = (v - \kappa)\|w\|_2^2 + (\kappa + \mu^2)(\mathbf{1}^T w)^2. \tag{2.12}$$

This separates coherent bias from covariance. For an unweighted sum, $\mathbb{E}[E_{\mathbf{1}}^2] = nv\mathcal{I}_n + n^2\mu^2$; an inflation factor alone therefore does not measure its mean-square error. For $\mathbf{1}^T w = 0$, the bias and the common-direction contribution vanish.

The effect of deleting $C$ depends on the weights. If $V_w$ denotes the true centered variance and $\widetilde{V}_w$ is obtained by replacing $\kappa = B + C$ by $B$ while retaining $v$, then

$$\widetilde{V}_w - V_w = -C\big((\mathbf{1}^T w)^2 - \|w\|_2^2\big), \qquad |\widetilde{V}_w - V_w| \le \frac{A}{n-1}\big|(\mathbf{1}^T w)^2 - \|w\|_2^2\big| \le A\|w\|_2^2. \tag{2.13}$$

Indeed, Corollary 2.2 gives $0 \le -C \le A/(n-1)$, and Cauchy–Schwarz gives the last inequality. The same difference applies to mean-square error because the retained bias is unchanged. For weights summing to zero, deleting $C$ underestimates the centered variance; for an unweighted sum it overestimates it.

PROPOSITION 2.3 (Symmetry and stochastic-rounding controls). *For the quantizer in (2.1), a distribution symmetric about zero gives $m_k = c_k = 0$ for every state, hence $\mu = \kappa = 0$ and $\mathcal{I}_n = 1$ whenever $v > 0$. For stochastic rounding with independent*

*rounding choices conditional on the full input block, $\mu = \kappa = 0$ for any input law, even without independence between the inputs. Its marginal variance is*

$$v_{\mathrm{SR}} = \mathbb{E}[s_{K_n}^2 \theta_1(1-\theta_1)], \qquad \theta_1 = \mathrm{frac}(X_1/s_{K_n}), \tag{2.14}$$

*with the integrand zero in the all-zero state.*

*Proof.* Oddness of round-to-nearest on symmetric truncation sets gives $T_{1,k}(t) = 0$, and Theorem 2.1 proves the first claim. Here stochastic rounding means ideal independent random bits conditional on the complete input block; finite-bit implementations require a separate randomness audit. Stochastic rounding has conditional mean zero and second moment $s^2\theta_i(1-\theta_i)$ [2, 3]. Independent choices give zero conditional cross moments; taking expectations proves (2.14). ∎

*Example* 2.4 (Conditional exclusion can produce variance deflation). Let $\mathbb{P}(X = 13/10) = \mathbb{P}(X = 41/20) = 1/2$ and $L = 2$. An all-small block has step $1/4$ and every error is $-1/20$. Every other block has step $1/2$; its small inputs have error $1/5$, and its large inputs have error $-1/20$. Direct enumeration gives

$$\mathcal{I}_n = 1 - \frac{(n-1)4^{-n}}{1/4 - 4^{-n}} < 1. \tag{2.15}$$

At $n = 2$, $v = 3/256$, $\kappa = -1/256$, and $\mathcal{I}_2 = 2/3$. The conditional-mean contribution is $B = 1/768$, so omitting $C$ instead predicts $10/9$. This reverses the qualitative conclusion.

### 2.3. Independent inputs with different distributions.

Proposition 2.5 (Conditional moments for independent nonidentical inputs). *Let $X_1, \dots, X_n$ be independent, each with a finite second moment, and use the same maximum-selected grid as in* (2.1). *At a fixed exponent $k$ with $P_k > 0$ (and hence $q_i > 0$ for every $i$), write $p_i = \mathbb{P}(|X_i| < a_k)$, $q_i = \mathbb{P}(|X_i| < b_k)$, and $P_k = \prod_i q_i - \prod_i p_i > 0$. For $\ell \in \{1, 2\}$ define*

$$z_{\ell,i} = \mathbb{E}[h_k(X_i)^\ell \mid |X_i| < b_k], \qquad u_{\ell,i} = \mathbb{E}[h_k(X_i)^\ell \mid |X_i| < a_k] \quad \text{when } p_i > 0.$$

*If every $p_i > 0$, put $r = \prod_i (p_i/q_i)$. Then*

$$\mathbb{E}[e_i^\ell \mid K_n = k] = \frac{z_{\ell,i} - r u_{\ell,i}}{1-r}, \tag{2.16}$$

$$\mathbb{E}[e_i e_j \mid K_n = k] = \frac{z_{1,i} z_{1,j} - r u_{1,i} u_{1,j}}{1-r}, \qquad i \neq j, \tag{2.17}$$

$$\mathrm{Cov}(e_i, e_j \mid K_n = k) = -\frac{r}{(1-r)^2}(z_{1,i} - u_{1,i})(z_{1,j} - u_{1,j}). \tag{2.18}$$

*If any $p_i = 0$, the conditional inputs are independent under the upper truncations: the marginal moments are $z_{\ell,i}$, the pair moments are $z_{1,i} z_{1,j}$, and all conditional pair covariances are zero. The all-zero state has probability $\prod_i \mathbb{P}(X_i = 0)$ and zero error moments.*

*Proof.* On the upper rectangular event $\{M_n < b_k\}$, each marginal moment factors as $\prod_i q_i$ times its normalized truncated moment; pair moments factor as $\prod_i q_i$ times the product of the two truncated means. Subtract the corresponding lower-event expressions and divide by $P_k$. Subtracting the product of the conditional means yields (2.18). If any $p_i = 0$, the lower event has probability zero, leaving only independent upper truncations. ∎

The unconditional moments follow by mixing these states, including the all-zero state. In particular, with $m_{i,k} = \mathbb{E}[e_i \mid K_n = k]$ and $c_{ij,k} = \mathrm{Cov}(e_i, e_j \mid K_n = k)$,

$$\mathrm{Cov}(e_i, e_j) = \mathrm{Cov}(m_{i,K_n}, m_{j,K_n}) + \sum_k P_k c_{ij,k}.$$

The conditional covariance can have either sign because the two truncated-mean differences need not agree in sign.

COROLLARY 2.6 (Conditional mixtures). *Suppose the coordinates are independent conditional on a random variable $Y$, and have finite unconditional second moments. Let $m(y)$ and $\Sigma(y)$ be the full error mean and covariance obtained from Proposition 2.5 at $Y = y$. Then*

$$\mathbb{E}[e] = \mathbb{E}_Y[m(Y)], \qquad \mathrm{Cov}(e) = \mathbb{E}_Y[\Sigma(Y)] + \mathrm{Cov}_Y(m(Y)). \tag{2.19}$$

*For several blocks, $\Sigma(y)$ is block diagonal; the second term includes covariances between blocks.*

*Proof.* Apply the conditional moment formulas for almost every $y$, followed by total expectation and total covariance. Since $|e_i| \le |X_i|$, the required moments are integrable. ∎

**3. Asymptotic behavior and an unbounded counterexample.** Bounded support fixes the exponent eventually. Unbounded support forces the grid to become arbitrarily coarse. The second mechanism removes correlation between a fixed pair of errors, but does not guarantee that the accumulated variance ratio tends to one.

PROPOSITION 3.1 (Bounded support and fixed-pair decorrelation). *The following statements hold at fixed $L$.*

1. *Suppose $|X|$ is bounded and $k_* = \max\{k : \mathbb{P}(|X| \ge 2^k) > 0\}$ exists. Put $p = \mathbb{P}(|X| < 2^{k_*}) < 1$ and $v_* = \mathrm{Var}(h_{k_*}(X))$. If $v_* > 0$, then*

$$v \longrightarrow v_*, \qquad |\kappa| = O(p^n), \qquad |\mathcal{I}_n - 1| = O(np^n). \tag{3.1}$$

   *If $p = 0$, the exponent is fixed and $\mathcal{I}_n = 1$ for every $n$.*
2. *Suppose $|X|$ has unbounded support and $\sigma_X^2 = \mathrm{Var}(X) > 0$. Then*

$$\begin{gathered} e_i \longrightarrow -X_i \text{ in } L^2 \text{ for each fixed } i, \qquad \mu \longrightarrow -\mathbb{E}[X], \\ v \longrightarrow \sigma_X^2, \qquad \kappa \longrightarrow 0, \qquad \frac{\mathcal{I}_n}{n} \longrightarrow 0. \end{gathered} \tag{3.2}$$

*Proof.* For bounded support, $\mathbb{P}(K_n \ne k_*) = p^n$. Couple $e_i$ to the fixed-step error $e_i^* = h_{k_*}(X_i)$ on the same inputs. These errors coincide outside that event and are uniformly bounded. Consequently, their first, second, and pair moments differ by $O(p^n)$. The $e_i^*$ are independent, so $\kappa = O(p^n)$, and $v \to v_* > 0$ gives (3.1).

For unbounded support, realize all blocks as prefixes of an infinite i.i.d. sequence. Almost surely $M_n \to \infty$, hence $s_{K_n} \to \infty$. For each fixed $i$, eventually $|X_i| < s_{K_n}/2$, and $\widehat{X}_i = 0$. The grid includes zero, so its nearest point satisfies $|e_i| \le |X_i|$ and $|\widehat{X}_i| \le 2|X_i|$. Dominated convergence gives $\widehat{X}_i \to 0$ in $L^2$. The limits of $\mu$, $v$, and $\kappa$ follow from this convergence and the independence of $X_1, X_2$. Equation (2.2) then gives $\mathcal{I}_n/n \to 0$. ∎

The unbounded result cannot be strengthened to $\mathcal{I}_n \to 1$ under these assumptions. The factor $n - 1$ in (2.2) can amplify a covariance that tends to zero. The following light-tailed example makes this obstruction explicit.

THEOREM 3.2 (Diverging inflation for exponential inputs). *Let $X \sim \mathrm{Exp}(1)$ and keep $L \geq 0$ fixed. For $a_j = 2^j$ and $n_j = \lfloor \exp(a_j) \rfloor$, define $\beta = 2^{-L-2}$. Then*

$$\mathcal{I}_{n_j} \sim e^{-1}(1-e^{-1})\, 2^{-2L-2} (\log n_j)^2 n_j^{1-2\beta} \qquad (j \to \infty). \tag{3.3}$$

*In particular $\mathcal{I}_{n_j} \to \infty$, although $\kappa_{n_j} \to 0$ and $\mathcal{I}_{n_j}/n_j \to 0$.*

*Proof.* Throughout the proof, $a = a_j$ and $n = n_j$. The two exponent probabilities are

$$P_{j-1} = (1-e^{-a})^n - (1-e^{-a/2})^n \longrightarrow e^{-1},$$
$$P_j = (1-e^{-2a})^n - (1-e^{-a})^n \longrightarrow 1 - e^{-1},$$

because $ne^{-a} \to 1$, $ne^{-a/2} \to \infty$, and $ne^{-2a} \to 0$. We first compute the conditional moments exactly, then control the remaining exponent states to obtain the asymptotic equivalence.

For a fixed step $s$, the layer-cake formula for the nonnegative quantized value gives

$$H(s) := \mathbb{E}[s\,\mathrm{RN}(X/s)] = s \sum_{m=1}^{\infty} e^{-(m-1/2)s} = \frac{s e^{-s/2}}{1-e^{-s}}, \qquad \mathbb{E}[h_s(X)] = H(s) - 1. \tag{3.4}$$

Here ties have probability zero. Away from ties, $h_s$ and $h_s^2$ are periodic with period $s$. For a positive integer $m$ and either response $\varphi = h_s$ or $h_s^2$, shifting the exponential tail gives

$$\mathbb{E}[\varphi(X)\mathbf{1}_{\{X \geq ms\}}] = e^{-ms}\mathbb{E}[\varphi(X)].$$

Both binade endpoints are integer multiples of their step: $a_k/s_k = 2^L$, $b_k/s_k = 2^{L+1}$. Consequently, truncation at either endpoint preserves the normalized first and second response moments. Theorem 2.1 therefore gives, for every state, $m_k = H(s_k) - 1$ and $c_k = 0$. The second moment follows by integrating one period:

$$t_k = \frac{\int_0^{s_k/2} x^2 e^{-x}\,dx + \int_{s_k/2}^{s_k} (s_k - x)^2 e^{-x}\,dx}{1 - e^{-s_k}} = 2 - 2H(s_k).$$

Thus, with $R = H(s_{K_n})$, the population moments have the closed form

$$\mu = \mathbb{E}[R] - 1, \qquad v = 1 - (\mathbb{E}[R])^2, \qquad B = \mathrm{Var}(R), \qquad C = 0. \tag{3.5}$$

For state $j-1$, the step is $s_- = a/2^{L+1}$; the step in state $j$ is $s_+ = 2s_-$. Put $H_- = H(s_-)$ and $H_+ = H(s_+)$. By (3.4), $H_- \sim 2^{-L-1} a e^{-\beta a}$ and $H_+/H_- \to 0$. Also $H(s) = s/(2\sinh(s/2))$ is decreasing and satisfies $0 < H(s) \leq 1$; the explicit hyperbolic form gives $H'(s) < 0$ and $H_+/H_- \to 0$. The derivative is negative for $s > 0$, which also shows $H_+/H_-$ tends to zero. The total probability of all lower states is

$$\delta_n = \mathbb{P}(K_n \leq j-2) = (1-e^{-a/2})^n \leq \exp(-ne^{-a/2}) = \exp(-\sqrt{n}(1+o(1))).$$

It follows that $\delta_n = o(H_-^2)$. In every upper state $k \geq j$, $R \leq H_+ = o(H_-)$. These bounds control all omitted states: the lower states contribute at most $\delta_n$ to either the first or the second moment of $R$, and the upper states contribute at most $H_+$ and $H_+^2$, respectively. Therefore

$$\mathbb{E}[R] = P_{j-1} H_- + o(H_-), \qquad \mathbb{E}[R^2] = P_{j-1} H_-^2 + o(H_-^2),$$

and hence $B \sim e^{-1}(1-e^{-1})H_-^2$. Equation (3.5) gives $v \to 1$ and $\mathcal{I}_n = 1+(n-1)B/v$. Since $1-2\beta > 0$, the latter expression is asymptotic to $nB$. Finally $a/\log n \to 1$ and $e^{-\beta a} \sim n^{-\beta}$, proving (3.3). ∎

For $L = 2$ the exponent in this asymptotic law is $7/8$. Thus a finite second moment guarantees fixed-pair decorrelation but allows the inflation of a block sum to grow without bound. The position of the block maximum relative to powers of two matters even for an exponential tail.

## 4. Computing the moments and designing the experiments.

**4.1. A deterministic evaluator.** The statistical moment formulas are exact; their implementation evaluates the required input integrals numerically. For a density $f$ and rounding cell $J_{d,k} = [(d-1/2)s_k, (d+1/2)s_k)$, the continuous contribution is

$$T_{\ell,k}(t) = \sum_d \int_{J_{d,k}\cap(-t,t)} (ds_k - x)^\ell f(x)\,dx. \tag{4.1}$$

Atoms contribute their probabilities times $h_k(x)^\ell$, with ties to even. Expanding the integrand uses cell probabilities and the first two truncated input moments. Since $t/s_k \le 2^{L+1}$ at both binade endpoints, each exponent requires $O(2^L)$ cells, independently of $n$. For heterogeneous inputs, these integrals are evaluated coordinatewise. The exponential identity (3.5) supplies a separate closed-form check.

We form $G(t)^n$ from survival probabilities and logarithms, and use `expm1` differences for nearby log probabilities. Equation (2.7) evaluates the conditional covariance stably; centered sums across exponent states evaluate $B$. The evaluator records omitted exponent probability and numerical stability diagnostics. The large-block exponential calculation uses higher precision to resolve survival probabilities below binary64 resolution.

We compare four predictions. The *finite-block prediction* uses (2.8)–(2.9). The *conditional-mean upper prediction* retains $m_k, t_k, v$ and sets $C = 0$, as in Corollary 2.2. The *unconditioned-response approximation* mixes moments of $h_k(X)$ under the original input law with $P_k$ and assumes conditional independence. The *uniform-noise prediction* uses $\mu = \kappa = 0$ and $v = \mathbb{E}[s_{K_n}^2]/12$.

For an iid law, the evaluator visits the retained exponent states and integrates $O(2^L)$ rounding cells per state, so its scalar work is $O(K2^L)$ for $K$ retained states and does not grow with the block length $n$. For a heterogeneous block of size $b$, the corresponding work is $O(K(b2^L + b^2))$ including covariance assembly; storage is $O(b^2)$. This is the cost of the population evaluator, separate from Monte Carlo sampling and from the linear solve used in the application experiments.

In the tables, "Exact" denotes numerical evaluation of the exact moment identities. The heterogeneous and latent evaluators retain exponents $-40 \le k \le 35$ and skip states of probability below $10^{-25}$. They record omitted-probability and residual-energy bounds separately from integration convergence checks. Numerical truncation bounds are therefore part of the population prediction, rather than evidence supplied by its Monte Carlo interval. The application codecs use an 8-bit signed exponent and reject out-of-range states; all retained source vectors and all population states lie inside that range, so no saturation or exponent clipping is included in the reported errors.

**4.2. Independent verification.** An independent rational oracle enumerates 25,600 blocks in 96 configurations: six finite-support laws, $n \in \{2,3,4,6\}$, and $L \in \{0,1,2,4\}$. It checks the conditional moments, covariance eigenvalues, weighted MSE,

and certified bracket, including atoms, zero blocks, dyadic endpoints, and ties to even. A second oracle enumerates 480 blocks in 32 heterogeneous-input configurations and checks Proposition 2.5, including both conditional covariance signs. Rational Green-matrix calculations verify linear propagation against direct gradient energy in one and two dimensions. Independent lognormal evaluators use analytical cell moments at 60 and 90 decimal digits to check the homogeneous and heterogeneous moment calculations. An 80-digit Pareto cell integrator agrees with all eight binary64 inflation factors to relative error below $8.1 \times 10^{-10}$.

**4.3. Sampling and measurement.** The main-grid and weighted experiments use independent blocks, master seed 20261010, and labeled child random streams. Population predictions are computed from the input law. For each sampled block we retain $S_b = \sum_i e_{b,i}$, $Q_b = \sum_i e_{b,i}^2$, exponent state, and squared step, with $s_* = 0$. For $B_{\mathrm{MC}}$ blocks,

$$\widehat{\mu} = \frac{\sum_b S_b}{nB_{\mathrm{MC}}}, \quad \widehat{v} = \frac{\sum_b Q_b}{nB_{\mathrm{MC}}} - \widehat{\mu}^2, \quad \widehat{\mathcal{I}}_n = \frac{\mathrm{sampleVar}_b(S_b)}{n\widehat{v}}. \tag{4.2}$$

The numerator has denominator $B_{\mathrm{MC}} - 1$. Confidence intervals are nominal 95% percentile intervals from 200 bootstrap resamples of whole blocks. We use "contains" below to report whether a displayed interval contains the population prediction; this is an empirical interval check, not a repeated-sampling coverage theorem. Application-specific streams, sample counts, and vector-bootstrap protocols are given with their experiments.

The main grid uses $\mathcal{N}(0,1)$, $\exp(Z)$, $|Z|$, $\max(Z,0)$ for $Z \sim \mathcal{N}(0,1)$, Exp(1), and Pareto(4) with $\mathbb{P}(X > x) = x^{-4}$ for $x \geq 1$. Combining $n \in \{32, 128, 512, 4096\}$ with $L \in \{2, 4\}$ gives 48 round-to-nearest configurations, with respectively 100,000, 40,000, 20,000, and 6,000 blocks. Eight stochastic-rounding controls use $n = 128$, both precisions, and Gaussian, lognormal, half-normal, and ReLU inputs. Six small-block cases use Beta$(2, 5)$ and exponential inputs at $n = 2, 4, 8$, $L = 2$, with 150,000 blocks each; two more Beta cases use $n = 32, 128$, $L = 2$, and the corresponding main-grid block counts. This gives 64 configurations.

Pareto(4) has an infinite fourth moment, and ordinary samples can miss rare exponent states that contribute substantially to second moments. We therefore sample all eight Pareto configurations by exponent strata and retain the ordinary results in the package. State $k$ receives $\max(100, \mathrm{round}(30{,}000P_k))$ conditional blocks. To draw one, sample the maximum by inversion uniformly between $G(a_k)^n$ and $G(b_k)^n$, then draw the remaining $n - 1$ coordinates independently below that maximum. Placing the maximum last preserves the symmetric statistics $S_b, Q_b$.

The stratified estimator combines the first and second moments of $S_b, Q_b$ with analytical state probabilities, renormalized over retained states. A variance correction removes the squared-mean estimation bias. Bootstrap resampling occurs within each stratum with its population weight fixed. Allocations and omitted-mass diagnostics are saved. For Pareto(4) and $T \geq 1$, the omitted upper-tail marginal second moment satisfies

$$\mathbb{E}[e_1^2 \mathbf{1}_{\{M_n \geq T\}}] \leq \frac{\mathbb{E}[M_n^2 \mathbf{1}_{\{M_n \geq T\}}]}{4\, 2^{2L}} \leq \frac{n}{2\, 2^{2L} T^2}. \tag{4.3}$$

Weighted reductions use 50,000 paired lognormal blocks at $n = 128$, $L = 2$, deterministic coefficients, and binary64 products and reductions. We record mean error, centered variance, and MSE separately for shared scales and partitions. The

TABLE 5.1

*New finite-n validation at $L = 2$, $n = 4096$. Monte Carlo brackets are 95% block-bootstrap intervals; Pareto uses conditional stratification. $\mathcal{I}_{\text{drop}}$ drops only $c_k$ while retaining exact conditional one-element moments; $\mathcal{I}_{\text{red}}$ additionally replaces those moments by unconditional responses.*

| Distribution | MC [95% CI] | Exact | $\mathcal{I}_{\text{drop}}$ | $\mathcal{I}_{\text{red}}$ |
|---|---|---|---|---|
| Gaussian | 1.013 [0.965, 1.056] | 1.000 | 1.000 | 1.000 |
| Half-normal | 14.86 [14.50, 15.19] | 14.927 | 14.927 | 14.905 |
| ReLU | 6.202 [5.913, 6.483] | 6.205 | 6.205 | 6.186 |
| Lognormal | 173.8 [168.0, 178.8] | 177.1 | 177.1 | 177.3 |
| Pareto(4) | 2893.7 [2893.0, 2894.7] | 2894.1 | 2894.1 | 2895.0 |
| Exponential | 41.78 [39.99, 43.73] | 41.794 | 41.794 | 41.794 |

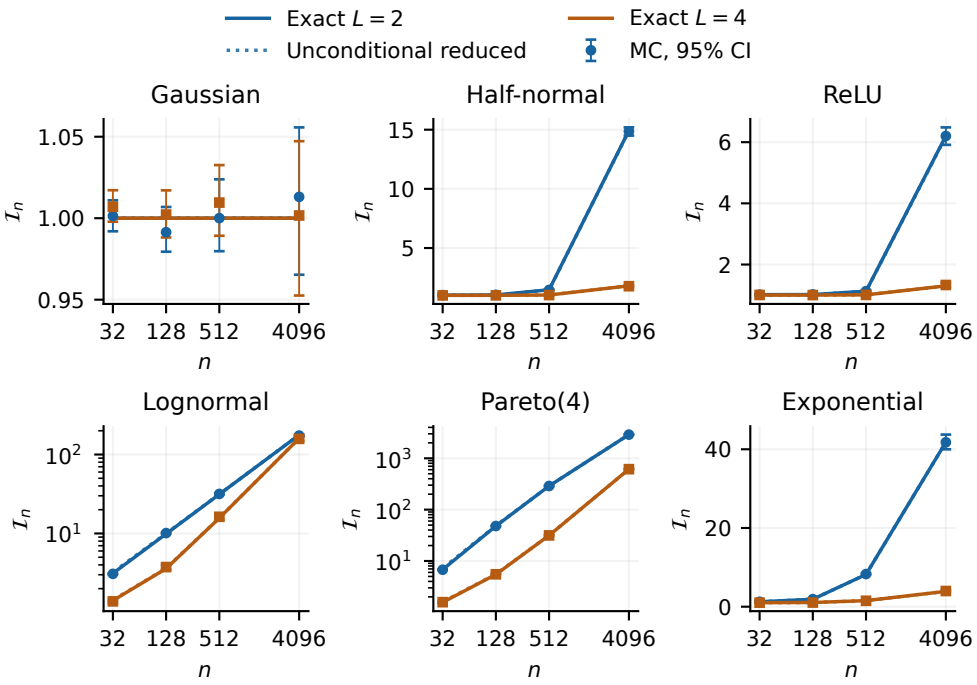



FIG. 5.1. *Finite-block validation on six input laws. Solid curves are exact moment predictions; dotted curves use unconditional responses and drop conditional covariance. Markers show Monte Carlo estimates with 95% intervals, using conditional stratification for Pareto.*

exponential asymptotic experiment evaluates the moment formulas at high precision along the theorem's subsequence.

## 5. Numerical results.

**5.1. Finite-block predictions and approximation errors.** Figure 5.1 and Table 5.1 compare the population calculation with independent block samples. Across 64 configurations, 61 nominal 95% intervals contain the predicted inflation, and the largest point discrepancy is 4.66%. The three misses are lognormal $(n, L) = (128, 2)$, exponential $(8, 2)$, and Beta$(2, 5)$ $(32, 2)$.

For asymmetric laws at $n = 4096$, $L = 2$, the predicted inflation is 6.205 for

TABLE 5.2

*Weighted dot-product quantization MSE divided by $\sum_i w_i^2$, lognormal inputs, $n = 128$, $L = 2$, 50,000 blocks. Brackets are 95% intervals. The weights are fixed before sampling. Balanced: alternating $1, -1$; near cancellation: alternating $1, -0.9$; positive: all ones.*

| Method | Balanced | Near cancellation | Positive |
|---|---|---|---|
| Shared RNE | 0.509 [0.500, 0.517] | 0.545 [0.535, 0.555] | 12.58 [12.40, 12.80] |
| Split blocks of 16 | 0.165 [0.162, 0.167] | 0.166 [0.164, 0.168] | 0.816 [0.805, 0.827] |
| Independent scales | 0.020 [0.020, 0.021] | 0.020 [0.020, 0.021] | 0.023 [0.023, 0.024] |
| Shared SR | 1.375 [1.345, 1.407] | 1.374 [1.348, 1.404] | 1.380 [1.350, 1.413] |

ReLU, 14.927 for half-normal, 177.1 for lognormal, 41.794 for exponential, and 2894.1 for Pareto(4). Changing the precision changes which binades place most inputs in the zero cell, producing distinct block-length trends. The conditional-mean upper prediction has maximum additive excess 0.00648 in the continuous-law grid. The unconditioned-response approximation instead has maximum relative discrepancy 5.81%, at Pareto(4), $n = 32$, $L = 2$: exact inflation is 6.806, the conditional-mean upper value is 6.812, and the unconditioned-response value is 7.201. The dominant approximation error here is in the conditional marginal moments.

Pareto inputs expose the importance of rare scale states. Ordinary sampling differs from the population prediction by as much as 14.8%; sampling every retained exponent stratum brings all eight predictions within their intervals. The omitted marginal second-moment bound is $3.0 \times 10^{-8}$ at $L = 2$ and $1.9 \times 10^{-9}$ at $L = 4$. Both sampling records are retained.

*Rounding controls..* All eight stochastic-rounding inflation intervals contain one. At $n = 128$, $L = 2$, the Gaussian marginal-variance ratio $12\widehat{v}/\widehat{\mathbb{E}[s^2]}$ is approximately 1.000 for round-to-nearest and 2.000 for stochastic rounding. For ReLU, the atom at zero contributes zero error; the ratios are approximately $1/2$ and 1, respectively. These controls check the symmetric residual, conditional unbiasedness, and treatment of atoms.

**5.2. Weighted reductions and block partition.** For $n = 128$, $L = 2$, we use positive weights, alternating $1, -1$, and alternating $1, -0.9$. Define

$$\chi = \frac{(\mathbf{1}^T w)^2}{n\|w\|_2^2} \in [0, 1]. \tag{5.1}$$

Equation (2.12) gives $\mathbb{E}[E_w^2]/\|w\|_2^2 = (v-\kappa)+n(\kappa+\mu^2)\chi$. The three vectors have $\chi = 1$, 0, and approximately 0.00276. Figure 5.2 and Table 5.2 compare shared-scale rounding, split blocks, and independently selected scales. The predictions have maximum MSE discrepancy 1.48%; eleven of twelve nominal intervals contain them.

Shared round-to-nearest gives normalized MSE 12.58 for positive weights and 0.509 for balanced weights. For positive weights, centered variance is about 695.1 and MSE is 1610.0, showing the additional squared-bias contribution. Stochastic rounding gives normalized MSE near 1.38 for both patterns: eliminating bias and covariance improves the positive reduction, while its larger marginal variance worsens the balanced reduction. Splitting into length-16 blocks gives positive-weight MSE 0.816; an independent scale per coordinate gives 0.023. The following elliptic experiments account for the storage cost of those scale choices.

*Asymptotic verification..* Table 5.3 evaluates the subsequence in Theorem 3.2 at $L = 2$, $j = 3, \ldots, 8$. Cell integration at 100 and 160 digits and the closed exponential

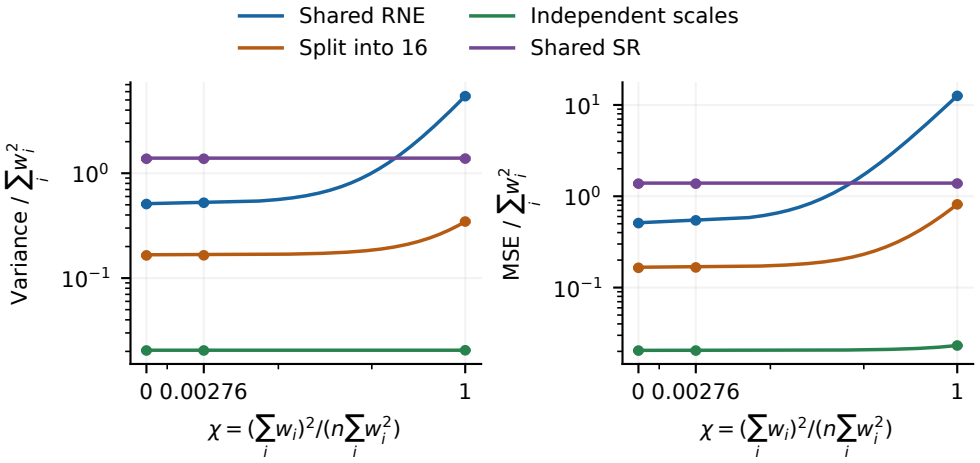



FIG. 5.2. *Weighted dot products for lognormal inputs ($n = 128$, $L = 2$). Fixed weights vary the normalized imbalance $\chi = (\sum_i w_i)^2/(n\sum_i w_i^2) \in [0, 1]$. Curves evaluate the affine laws from exact moments on a dense grid; markers show Monte Carlo with 95% intervals.*

TABLE 5.3

*Exact high-precision exponential counterexample at $L = 2$, $n = \lfloor e^{2^j} \rfloor$. The theorem normalization is $C(\log n)^2 n^{7/8}$ with $C = e^{-1}(1 - e^{-1})/64$. Convergence of the ratio to one is predicted by the asymptotic theorem. All displayed values agree between 100 and 160 decimal digits.*

| $2^j$ | $\mathcal{I}_n$ | $\mathcal{I}_n/n$ | Exact / normalization |
|---|---|---|---|
| 8 | $4.111 \times 10^{1}$ | $1.380 \times 10^{-2}$ | 0.161276 |
| 16 | $3.296 \times 10^{5}$ | $3.709 \times 10^{-2}$ | 0.294655 |
| 32 | $3.298 \times 10^{12}$ | $4.177 \times 10^{-2}$ | 0.612870 |
| 64 | $2.901 \times 10^{25}$ | $4.652 \times 10^{-3}$ | 0.931787 |
| 128 | $2.601 \times 10^{50}$ | $6.690 \times 10^{-6}$ | 0.998663 |
| 256 | $4.558 \times 10^{99}$ | $3.016 \times 10^{-12}$ | 1.000000 |

formula agree in all displayed digits. Inflation grows from about 41 to $4.558\times10^{99}$, while the final $\mathcal{I}_n/n$ is $3.016\times10^{-12}$ and the ratio to the theorem normalization is 0.99999955. Here $\mathcal{I}_n$ is a centered variance ratio; it is not the absolute mean-square error, whose additional coherent-bias term is given in (2.12). The high-precision calculation resolves population-moment cancellation and does not claim that a binary64 reduction would form the displayed sum without overflow or loss of significance.

**5.3. Spatially heterogeneous sources in an elliptic solve.** Consider $-\Delta u = f$ on $(0, 1)^2$ with homogeneous Dirichlet conditions. On an $m \times m$ interior grid, let $h = (m + 1)^{-1}$, $A = h^{-2}(I \otimes T + T \otimes I)$ with $T = \mathrm{tridiag}(-1, 2, -1)$, and $G = A^{-1}$. We compress the cell sources $b_i = s_i \exp(Z_i)$, where the $Z_i$ are independent $N(0, 1)$ variables. The prescribed spatial scales are

$$s_{\mathrm{sm}}(x, y) = 1 + 3\sin^2(\pi x)\sin^2(\pi y),$$

$$s_{\mathrm{hot}}(x, y) = 1 + 7\exp\left(-\frac{(x - 0.3)^2 + (y - 0.65)^2}{0.02}\right).$$

These profiles test the nonidentical-input formula with smooth variation and a localized source hotspot.

TABLE 5.4

*Two-dimensional heterogeneous source compression, $32^2$ unknowns, contiguous blocks, and exactly 7168 payload bits. Source and Energy allocate the extra coefficient bits by source scale and the full energy prediction, respectively. Relative energy RMS values are percentages. Ind. retains exact marginal moments and bias; Unif. uses independent uniform-cell noise.*

| Profile | $\|B\|$ | Placement | Exact | Ind. | Unif. | MC 95% interval |
|---|---|---|---|---|---|---|
| Smooth | 8 | Uniform | 1.244 | 1.206 | 1.141 | [1.242,1.248] |
| Smooth | 32 | Uniform | 1.450 | 1.363 | 1.252 | [1.447,1.457] |
| Smooth | 32 | Source | 1.125 | 1.079 | 1.023 | [1.124,1.132] |
| Smooth | 32 | Energy | 1.125 | 1.079 | 1.023 | [1.124,1.131] |
| Smooth | 128 | Uniform | 3.071 | 2.541 | 1.962 | [3.058,3.108] |
| Smooth | 128 | Energy | 2.132 | 1.820 | 1.557 | [2.124,2.159] |
| Hotspot | 8 | Uniform | 1.471 | 1.423 | 1.358 | [1.468,1.481] |
| Hotspot | 32 | Uniform | 1.997 | 1.866 | 1.583 | [2.000,2.016] |
| Hotspot | 32 | Source | 1.543 | 1.451 | 1.282 | [1.544,1.556] |
| Hotspot | 32 | Energy | 1.529 | 1.439 | 1.278 | [1.530,1.542] |
| Hotspot | 128 | Uniform | 4.184 | 3.563 | 2.361 | [4.182,4.233] |
| Hotspot | 128 | Energy | 3.216 | 2.726 | 2.012 | [3.224,3.262] |

Write $e = Q(b) - b$, $\delta u = Ge$, $\mu = \mathbb{E}e$, and $\Sigma_B = \mathrm{Cov}(e_B)$ for a partition into disjoint blocks. For any symmetric positive definite $A$, define $\|z\|_{A,h}^2 = h^2 z^T A z$. Independence across blocks gives

$$J := \mathbb{E}\|\delta u\|_{A,h}^2 = h^2 \left[ \sum_B \mathrm{tr}(G_{BB}\Sigma_B) + \mu^T G \mu \right]. \tag{5.2}$$

The bias term couples different spatial blocks. Relative RMS is $\sqrt{J/\{h^2\mathbb{E}[b^T G b]\}}$, with the denominator evaluated from the source laws. The baseline denoted Ind. retains every exact marginal mean and variance while dropping off-diagonal error covariances; Unif. uses independent zero-mean uniform-cell errors.

All codecs pack exactly $7m^2$ payload bits: an 8-bit signed exponent per block, $L_B + 3$ signed bits per coefficient, and any remaining padding. Block lengths are $8, 32, 128$, with contiguous and interleaved layouts. The largest common precision is supplemented by prescribed extra-bit blocks. Their locations are fixed codec maps, so no data-dependent flags are needed. At $m = 32$, we compare evenly spaced locations (Uniform), descending $\sum_{i\in B} s_i^2$ (Source), and a greedy full-energy rule (Energy). For a candidate precision increment, embed its mean change $d$ in the full vector and let $\Delta\Sigma_B$ be its block covariance change. The greedy rule chooses the smallest

$$\Delta J = h^2 \left[ \mathrm{tr}(G_{BB}\Delta\Sigma_B) + 2d^T G\mu + d^T G d \right]. \tag{5.3}$$

Ties are resolved by block index. The implemented codec rejects out-of-range exponents; none occurred, and the per-vector probability bound is below $10^{-1632}$.

We retain all 40 prescribed configurations, including coincident precision maps, and use 20 000 paired source vectors per profile for $m = 16$ and 60 000 for $m = 32$, with 300 whole-vector bootstrap replicates. The latter pools the initial 10 000 vectors and a declared independent 50 000-vector replication for both profiles. All streams, configuration records, and interval misses remain in the package. The primary nominal intervals contain 34 of 40 population RMS predictions; the largest RMS discrepancy is 0.83%.

For the smooth profile at $m = 32$, length-32 Energy placement reduces measured RMS by 22.37% relative to Uniform placement (paired 95% interval $[22.17, 22.61]$%).

TABLE 5.5
*Correlated hotspot sources: relative energy RMS (percent).*

| $\lvert B_j\rvert$ | Layout | Full | Diagonal | MC 95% interval |
|---|---|---|---|---|
| 8 | Cont. | 2.406 | 2.335 | [2.370,2.428] |
| 8 | Inter. | 2.484 | 2.428 | [2.458,2.509] |
| 32 | Cont. | 3.124 | 2.832 | [3.068,3.154] |
| 32 | Inter. | 3.072 | 2.867 | [3.048,3.119] |
| 128 | Cont. | 9.536 | 7.940 | [9.385,9.721] |
| 128 | Inter. | 8.171 | 6.989 | [8.002,8.218] |

For the hotspot, it improves on Source placement by 0.86% ($[0.81, 0.91]\%$). These are gains from the full-model placement objective. The uniform-cell model selects length 32 with Energy placement for the hotspot, whereas the exact and marginal-moment models select length 8. The former choice has 4.12% higher measured RMS ($[3.71, 4.48]\%$), so the population model also supports a concrete accuracy decision at the same storage budget. The largest covariance gap occurs for the $16^2$ smooth case with length-128 contiguous blocks: exact, Ind., and measured RMS are 5.845%, 4.769%, and 5.856%, respectively; the diagonal-moment model underestimates energy MSE by a factor 1.50.

For each codec, 100 paired original and compressed sources are also solved by conjugate gradients at relative tolerance $10^{-11}$. For $r = A\widehat{\delta u} - e$, the solver discrepancy satisfies $\|\widehat{\delta u} - \delta u\|_{A,h} \le h\|r\|_2/\sqrt{\lambda_{\min}(A)}$; its maximum relative certificate is $2.8\times10^{-9}$, and the CG and spectral energy values differ by at most $1.3 \times 10^{-10}$ relatively.

**5.4. A correlated source field.** For correlated inputs on the $16 \times 16$ grid, use the same hotspot envelope $a_i$ and let

$$b_i = a_i \exp(Z_i + 0.6\phi_i Y), \qquad \phi_i = \sin(\pi x_i)\sin(\pi y_i),$$

where $Y, Z_1, \dots, Z_N$ are independent standard normals. Corollary 2.6 gives the energy prediction

$$\mathbb{E}\|\widehat{u} - u\|_{A,h}^2 = h^2 \mathbb{E}_Y\left[\operatorname{tr}(G\Sigma(Y)) + m(Y)^T G m(Y)\right]. \tag{5.4}$$

The diagonal comparison replaces $\Sigma(Y)$ by its diagonal while retaining the entire second term, including the correlations induced by the latent factor across distinct blocks. The reference source energy is analytical: for $M_i = a_i \exp((1 + 0.36\phi_i^2)/2)$, $\mathbb{E}[b_i b_j] = M_i M_j \exp(0.36\phi_i\phi_j)$ for $i \ne j$, and $\mathbb{E}[b_i^2] = a_i^2 \exp(2(1 + 0.36\phi_i^2))$.

All six prescribed block/layout codecs use $7N$ bits. Gauss–Hermite orders 21, 41, 61 give a final relative change of $2.5 \times 10^{-13}$ in energy MSE and the conditional-mean contribution. A new stream of 30 000 source vectors is paired across codecs, with 300 vector-bootstrap replicates. The measured nominal intervals contain all 6 population predictions. Actual CG solves on 100 vectors per codec have maximum relative energy difference $8.2 \times 10^{-11}$ from the direct source-error energies.

The diagonal model underestimates energy MSE by as much as a factor 1.44. This comparison isolates the shared-exponent contribution after the latent source correlation is included; "diagonal" here removes only conditional block off-diagonals and is not a fully independent-source model.

*Sensitivity to the common source factor..* For two fixed contiguous codecs ($|B| = 32, 128$), we also prescribe $\tau \in \{0, 0.3, 0.6, 0.9\}$ in $b_i = a_i \exp(Z_i + \tau\phi_i Y)$. Each setting uses 200 000 iid source vectors, paired across codecs, and 1000 whole-vector bootstrap

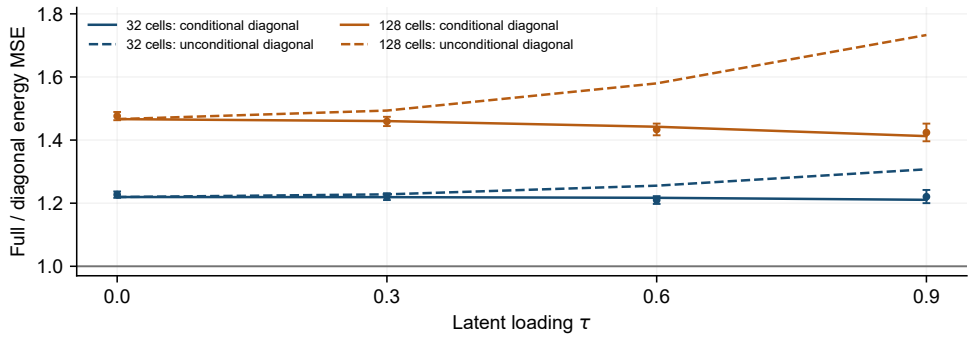



FIG. 5.3. *Common-factor sensitivity at seven bits per source value. Solid curves retain the full latent mean response in the diagonal comparison; dashed curves also remove unconditional off-diagonal covariances. Markers and nominal* 95% *intervals use independent source vectors.*

replicates. Gauss–Hermite orders 41 and 61 differ by at most $9.3 \times 10^{-9}$ in the full and comparison energy predictions. The nominal intervals contain 8 of eight population predictions, with maximum RMS discrepancy 0.41%. This is an empirical interval check rather than a repeated-sampling coverage claim; all cases are retained in the saved records. Figure 5.3 shows that removing conditional block covariances still underestimates MSE throughout this range. For length 128, the full/conditional-diagonal ratio changes from 1.47 to 1.41. Removing the latent mean covariances as well increases the ratio to 1.73 at $\tau = 0.9$. The two losses of covariance are thus distinct, even in this one-factor family.

**5.5. High-contrast diffusion and spatial bit placement.** We next use $-\nabla\cdot(\alpha\nabla u) = f$ on the same $16^2$ grid, where $\alpha = 0.01$ inside $(x-0.7)^2+(y-0.3)^2 \le 0.05$ and $\alpha = 1$ elsewhere. Harmonic face conductivities, including the Dirichlet boundary faces, define the symmetric positive definite operator. The low-conductivity region is separated from the source hotspot, so the compression objective accounts for spatial response as well as source scale. Using the independent hotspot source of Section 5.3 and 7 bits per cell, we retain all 14 prescribed codecs and 30 000 paired source vectors. At length 32, the full-energy rule changes two precision blocks relative to Source placement: it moves the extra-bit strip from $x \approx 0.06, 0.12$ toward the low-conductivity region at $x \approx 0.76, 0.82$ (Figure 5.4). This reduces measured RMS by 19.97% (paired 95% interval $[19.57, 20.37]$%); at length 128, the reduction is 11.25% ($[10.34, 12.17]$%). Table 5.6 gives the contiguous results; both layouts and coincident maps remain in the package. The nominal intervals contain 13 of 14 population RMS predictions, with maximum discrepancy 0.51%. We also solve 100 paired sources per codec by Jacobi-preconditioned conjugate gradients at tolerance $10^{-11}$. Since $A \succeq 0.01 A_{\mathrm{Poisson}}$, the preceding residual certificate uses $0.01\lambda_{\min}(A_{\mathrm{Poisson}})$; its largest relative bound is $8.4 \times 10^{-9}$, while PCG and direct-solve energy values agree within $4.7 \times 10^{-11}$ relatively.

**5.6. Calibration from a separate source sample.** The preceding high-contrast experiment assumes the source law is known. We now separate law fitting from codec validation on the same fixed $16^2$ operator. For each $T \in \{32, 128, 512, 2048\}$ and five independently labeled training samples, we fit $\log b_i = \ell_i + \sigma Z_i$ by coordinate-wise means and one pooled residual scale. The fitted $(\ell_i, \sigma)$ determine the source-scale and

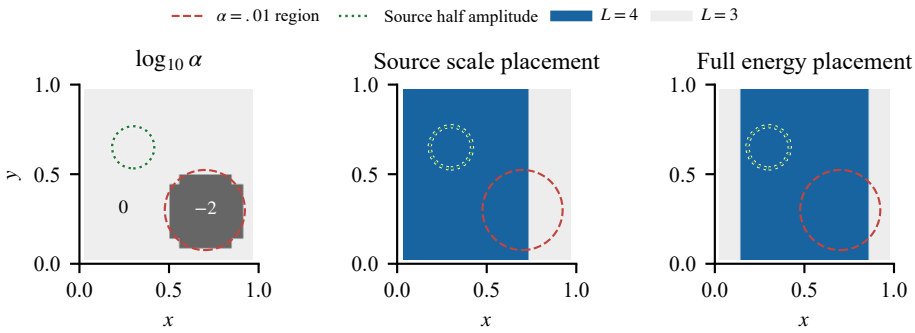



FIG. 5.4. *The prescribed diffusion coefficient and two length-32 contiguous precision maps. The red disk marks the low-conductivity region; the green outline is the half-amplitude source contour. Both maps have six blue $L = 4$ blocks and two gray $L = 3$ blocks at the same storage cost.*

TABLE 5.6
*High-contrast diffusion,* $16^2$ *sources and exactly* 1792 *payload bits. All entries use contiguous blocks; relative energy RMS is in percent.*

| $\|B\|$ | Placement | Exact | Ind. | MC 95% interval |
|---|---|---|---|---|
| 8 | Uniform | 3.213 | 3.159 | [3.202,3.243] |
| 32 | Uniform | 4.060 | 3.845 | [4.039,4.103] |
| 32 | Source | 3.887 | 3.735 | [3.870,3.925] |
| 32 | Energy | 3.114 | 3.005 | [3.099,3.139] |
| 128 | Source | 8.948 | 7.993 | [8.817,8.991] |
| 128 | Energy | 7.916 | 6.761 | [7.824,7.978] |

full-energy bit maps; each map is frozen before a common 30,000-vector validation stream is generated. The uniform map, the oracle map obtained from the known law, and every learned map are retained; the validation stream is never used to choose a map.

The full-moment energy map has median regret 0.00% at $T = 2048$ and 0.00% at $T = 32$; its central 95% replicate quantile range at $T = 32$ is $[0.00, 1.38]$%. The fitted source scale converges with $T$ (median log-scale RMSE 0.1714 at $T = 32$ versus 0.0222 at $T = 2048$). The median validation error of the train-only full-moment prediction is -5.43% at $T = 32$ and -1.01% at $T = 2048$. The fitted energy map recovered the oracle map in 17 of the 20 training fits; the other maps and their validation outcomes are retained. The validation stream evaluates six unique maps (43 candidate records, including repeated maps), and all 20 fits, 40 learned maps, predictions, validation energies, and bootstrap intervals are retained. The nominal intervals contain all six unique-map predictions; repeated candidate rows are bookkeeping records rather than independent validation tests. The validation stream is independent of every earlier experiment.

**6. Implications for reductions and compression.** The moment formulas connect the input distribution, block partition, and linear operator to the output error. For identical coordinate laws, the weight sum determines how strongly the common error component enters a reduction. Balanced weights remove the bias and the term proportional to $(\mathbf{1}^T w)^2$, leaving $(v - \kappa)\|w\|_2^2$. Positive weights can instead accumulate both covariance and bias. This explains why stochastic rounding can improve a

TABLE 5.7
*Validation RMS and regret for the learned 32-cell contiguous maps. Values are medians over five training replicates; regret is relative to the fixed oracle full-energy map, in percent.*

| $T$ | Source RMS | Energy RMS | Source regret | Energy regret |
|---|---|---|---|---|
| 32 | 4.498 | 3.100 | 45.07 | 0.00 |
| 128 | 3.956 | 3.100 | 27.58 | 0.00 |
| 512 | 3.956 | 3.100 | 27.58 | 0.00 |
| 2048 | 3.890 | 3.100 | 25.46 | 0.00 |

positive-weight reduction while increasing the mean-square error of a balanced one: its larger marginal variance is then the deciding term. The appropriate rounding rule depends on the weights and the accuracy criterion.

Partitioning changes both the selected scales and the covariance structure. Each candidate partition therefore requires its own moments. The storage cost of additional scales also changes the available coefficient precision. In an elliptic solve, block layout further determines how the covariance aligns with the Green operator. A model that retains only marginal error variances loses this interaction, even when it estimates those variances accurately. The equal-budget experiments demonstrate how the complete moments predict solution error and guide the comparison of prescribed compression choices. The calibration experiment shows how this workflow degrades when the source law is estimated from a separate sample.

Block length and total problem size have different effects. If $N = Kn$ independent inputs occupy $K$ identical blocks of fixed length $n$, their total error satisfies

$$\mathbb{E}\left[\left(\sum_{i=1}^{N} e_i\right)^2\right] = Nv\mathcal{I}_n + N^2\mu^2. \tag{6.1}$$

The centered standard deviation grows as $\sqrt{N}$, while a nonzero mean error contributes quadratically to mean-square error. An inflation factor describes the centered component; accuracy assessment requires both terms.

The exponential counterexample concerns growth of the block itself. At fixed grid precision, its increasing maximum makes each fixed coordinate eventually round to zero. The error approaches the negative input, and pair correlation vanishes, but multiplication by $n - 1$ still permits diverging inflation. The dyadic location of the maximum produces pronounced changes with block length. The exact finite-block evaluator resolves these changes without extrapolating a power law from one phase.

Latent factors separate a structured field's common fluctuations from its independent innovations. Conditional moment evaluation preserves dependence between spatial blocks through the varying mean error. More general joint laws require joint truncated moments; data-dependent weights require mixed moments of the weights and quantization errors. Quantizing both operands and including product or accumulation rounding adds further error terms. Saturation, nonuniform element grids, and finite random bits can be incorporated by supplying the corresponding scalar quantizer moments [4]. The applications use a uniform integer-mantissa grid with an 8-bit signed exponent and binary64 arithmetic. They establish accuracy comparisons at a fixed payload budget for specified source laws and discrete operators. Transfer to measured scientific fields, nonuniform MX element grids, saturation, and low-precision accumulation requires separate validation. Encoding cost, memory traffic, and solver

speed were not measured.

**7. Conclusions.** Maximum-selected block quantization creates two competing covariance terms: variation of the conditional error mean and the correction caused by excluding an all-small block. One-dimensional truncated moments compute both exactly and give variance and mean-square error predictions for weighted reductions. For identical input laws, dropping the conditional pair correction has a distribution-free additive error bound of one in the variance inflation factor. The heterogeneous and conditional-mixture extensions supply the covariance for spatially varying and correlated sources.

The support limits and exponential counterexample show why fixed-pair decorrelation does not imply unit inflation. The experiments verify the finite-block identities and connect the resulting moments to weight balance, storage allocation, spatial layout, and elliptic solution error. Together, the analysis and evaluator turn a statistical quantization model into a computable accuracy prediction for a specified input law and compression design.

**Data and code availability.** The accompanying reproduction package contains the experiment driver, saved machine-readable results, figure and table generator, high-precision evaluator, exact rational enumeration oracle, and a manifest with random seeds, package versions, execution times, and file hashes. The figures and tables in this manuscript are generated from that package. The existing project repository is https://github.com/NUAApanda-1116/bfp-error-independence; it records the project history, while the accompanying revision package is the submitted snapshot and specifies the files required to reproduce this version.

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